

Project Overview

The **Cybersecurity Dashboard** is a web-based application designed to monitor system and network logs, visualize security metrics, and provide real-time threat alerts. The project integrates a backend built with Flask, a frontend developed using React, and a database to store security logs.

Project Features

- **Real-time Log Monitoring:** Collects system logs, network logs, and other security-related data.
- **Data Visualization:** Displays CPU usage, disk usage, memory consumption, and other system metrics using interactive charts.
- **Threat Detection Alerts:** Provides notifications for potential security threats based on log analysis.
- **REST API Integration:** The backend exposes APIs to fetch and process log data.

What We Learned

1. **Log Collection:** Implemented a script to collect system logs and structured the data for storage.
2. **Backend Development:** Built a Flask-based API to serve log data to the frontend.
3. **Frontend Development:** Developed a React-based UI with Axios for API calls and Chart.js for data visualization.
4. **Integration & Deployment:** Learned how to connect frontend and backend, resolve compatibility issues, and deploy the application.
5. **Troubleshooting Errors:** Faced and resolved various errors related to module installations, Node.js versions, and Flask dependencies.

Next Steps

- Enhance the dashboard with user authentication.
- Implement advanced threat detection mechanisms.
- Optimize database performance for handling large log files.

This project serves as a foundational step towards building a comprehensive **SIEM (Security Information & Event Management) system**. 🚀

CyberGuard

Security Operations Center

Overview

Threats

Compliance

Assets

Security Dashboard

Last updated: Just now

Last 24 Hours

Generate Report

Active Threats

23

High Alert

Security Score

86%

Systems Protected

342

+12 New

Patch Status

94%

Up to date

Recent Security Events

System Health

CPU

72%

Memory

84%

Disk

85%

Network

91%